

Extractive vs. Abstractive Summarization of Long Documents: A Comparative Study with Demographic Bias Analysis

Layan Alfawzan Rose Mady Jana Alruzuq Layan Aldbays Aljawharah Alwabel

King Saud University

{444200793, 444200107, 444201157, 444200653, 444200750 }@student.ksu.edu.sa

Abstract

Evaluating trade-offs between extractive and abstractive summarization is fundamental to advancing long-document NLP. While extractive methods offer computational efficiency, they often fail to preserve narrative coherence in literary texts. To investigate this limitation, we compare a DistilBERT-based extractive baseline with three instruction-tuned LLMs—LLaMA 3, Mistral, and Qwen2—on the BookSum dataset. Additionally, we introduce a bias analysis framework with seven prompt conditions injecting demographic cues about author age, gender, and literacy.

Empirical results show that all three LLMs consistently outperform the extractive baseline. The extractive baseline produces fully copied summaries, whereas LLM outputs show very low sentence-level overlap with the reference summaries and appear predominantly abstractive. Regarding demographic bias, while some prompt conditions achieve statistical significance, effect sizes remain negligible across all models, indicating no practically meaningful demographic sensitivity.

These findings suggest that current instruction-tuned LLMs are robust to demographic framing in long-document summarization, underscoring the critical distinction between statistical and practical significance in bias evaluation.

1 Introduction

Automatic text summarization has emerged as a critical research direction in natural language processing (NLP), driven by the exponential growth of digital text and the practical need to efficiently extract key information from large documents. While short-document summarization has been extensively studied, summarizing long-form literary texts—such as full book chapters—remains a substantially more challenging problem. Long documents introduce issues of context retention, narrative coherence, and semantic compression that

push the limits of current models. The BookSum dataset (Krzyściński et al., 2021), which consists of book chapter–summary pairs drawn from multiple literary analysis sources, provides a rich and realistic benchmark for evaluating summarization systems at scale.

Text summarization approaches broadly fall into two paradigms: extractive and abstractive. Extractive methods, such as BERT-based sentence scoring (Devlin et al., 2019), select and concatenate the most salient sentences from the source document without generating new text. These approaches are computationally efficient and interpretable but are limited in their ability to produce coherent, fluent summaries, particularly for long documents with complex discourse structures. Abstractive methods, powered by large pre-trained language models (LLMs) such as LLaMA 3 (Dubey et al., 2024), Qwen2 (Hui et al., 2024), and Mistral (Jiang et al., 2023), generate novel text that may paraphrase, compress, or reorganize the source content. Recent advances in instruction-following LLMs have demonstrated strong abstractive summarization capabilities; however, their sensitivity to prompt framing—including demographic and stylistic cues—remains an open and underexplored question.

A growing body of work has documented the presence of demographic bias in LLMs, whereby model outputs vary systematically based on perceived attributes of users or subjects, including age, gender, and literacy level (Gallegos et al., 2024). Such biases, if present in summarization systems, could lead to inequitable information access and distorted representations of literary content. While bias has been studied in classification and generation tasks, its manifestation in long-document summarization—where the model is conditioned on both a lengthy source text and a demographic-laden prompt—has received limited attention.

This paper presents a systematic comparative

study of extractive and abstractive summarization methods on the BookSum dataset, with a dual focus on performance evaluation and demographic bias analysis. Specifically, we make the following contributions:

- We implement and evaluate an extractive baseline based on DistilBERT sentence similarity alongside three state-of-the-art abstractive LLMs (LLaMA 3-8B-Instruct, Qwen2-1.5B-Instruct, and Mistral-7B-Instruct) on 1,174 test samples.
- We introduce a controlled bias evaluation framework comprising seven prompt conditions that inject demographic cues related to author age, gender, and literacy into the summarization prompt.
- We apply both quantitative metrics (ROUGE-1, ROUGE-2, ROUGE-L, BERTScore) and rigorous statistical analysis (Wilcoxon signed-rank test, Cohen’s d effect size, 95% confidence intervals) to assess the practical significance of any observed performance differences.

To the best of our knowledge, this is the first study to simultaneously compare extractive and abstractive summarization paradigms on the BookSum benchmark while conducting a multi-model demographic bias evaluation under controlled prompt conditions. Unlike prior bias studies that focus on short-text classification or open-ended generation, our work examines bias in the context of long-document abstractive summarization, where source content is rich and the model’s reliance on prompt framing is less obvious.

Our experiments reveal that abstractive LLMs consistently outperform extractive baselines across ROUGE and BERTScore metrics. Critically, the bias analysis shows that none of the three LLMs exhibit meaningful sensitivity to demographic prompt cues: while some conditions yield statistically significant differences under the Wilcoxon test, effect sizes (Cohen’s d) remain negligible across all models and conditions, confirming that the observed variations are not practically significant.

The remainder of this paper is organized as follows. Section 2 reviews related work on text summarization and bias in LLMs. Section 3 describes the dataset, preprocessing pipeline, and summarization methods. Section 4 presents the experimental

results and statistical analysis. Section 5 discusses the findings and their implications. Section 6 concludes the paper.

2 Related Work

2.1 Long-Form Narrative Summarization and the BookSum Dataset

The BookSum dataset (Kryscinski et al., 2022), loaded in the present study via the kmfoda/booksum mirror on HuggingFace, is the primary benchmark establishing this regime. It contains chapter-summary pairs drawn from eight online sources—including Shmoop, GradeSaver, and SparkNotes—covering 170 unique books across training, validation, and test splits. Exploratory analysis conducted as part of the present study reveals that the median chapter length is 2,934 words and the mean is 4,021 words, with 96% of chapters exceeding BERT’s approximate 380-word input limit and 48.6% exceeding the LED model’s approximate 3,000-word limit. These figures confirm quantitatively that no single-pass encoder can process the majority of BookSum chapters without a chunking strategy, motivating the hierarchical sentence-aware chunking pipeline developed in this work.

A further complication identified during dataset analysis is the substantial variation in reference summary length across sources: GradeSaver produces summaries averaging 502 words, while BookWolf averages only 194 words—a 2.6-fold difference. This source-level variance means that ROUGE scores computed against BookSum references are sensitive to which source supplied the reference, a methodological consideration that must be accounted for when interpreting cross-model comparisons.

2.2 Extractive Summarization: Capabilities and Limitations

Extractive summarization methods select salient sentences from the source document rather than generating new text. The theoretical foundation for modern embedding-based extractive systems is Sentence-BERT (Reimers and Gurevych, 2019), which adapts the BERT encoder through siamese and triplet network training to produce sentence embeddings that are directly comparable via cosine similarity. This framework made embedding-based sentence selection computationally tractable, reducing pairwise semantic search from approximately

65 hours to around 5 seconds for 10,000 sentences.

The extractive baseline implemented in the present study uses `distilbert-base-uncased` (Devlin et al., 2019) to encode each candidate sentence independently. Specifically, for each sentence the model produces a 768-dimensional CLS token embedding, which serves as the sentence representation. A mean document embedding is then computed across all sentence embeddings, and sentences are ranked by cosine similarity to this centroid. The top three sentences, preserved in their original order, constitute the output summary. This approach draws directly on the theoretical properties established by Reimers and Gurevych (Reimers and Gurevych, 2019) and the distillation principles of Wang et al. (Wang et al., 2020).

However, the literature consistently identifies fundamental limitations of extractive centrality-based methods for narrative summarization. First, selecting semantically central sentences does not guarantee narrative coherence: a sentence may be lexically representative of the chapter without contributing to its plot arc. Second, extractive methods cannot paraphrase, compress information across multiple passages, or restructure discourse—capabilities that are essential when summarizing chapters where key events are distributed across thousands of words. Third, because the DistilBERT-based baseline encodes each sentence independently and the present baseline applies sentence-level truncation at 512 tokens, content appearing in the latter portions of long chapters may receive reduced representational coverage. These limitations are not incidental; they are structural properties of the centrality-based extractive paradigm, and they predict that the baseline will underperform generative LLMs on BookSum’s long-form narrative chapters—a hypothesis tested empirically in the present study.

2.3 Instruction-Tuned Large Language Models for Summarization

The three LLMs evaluated in this study represent three distinct approaches to improving decoder-only Transformers beyond architectural scale. Mistral-7B-Instruct-v0.1 (Jiang et al., 2023) introduces grouped-query attention (GQA) and sliding-window attention (SWA) to a 7.3-billion-parameter backbone, reducing key-value cache overhead while maintaining a large effective receptive field across layers. The base model achieves 60.1 on MMLU and 52.2 on GSM8K,

and the instruction-tuned variant scores 6.84 on MT-Bench—outperforming larger Llama 2 models. The architectural claim is that efficiency-oriented design, rather than scale alone, drives these gains.

Meta-Llama-3-8B-Instruct (Dubey et al., 2024) takes the opposite approach: a standard decoder backbone with a 128K-token vocabulary and GQA, trained on over 15 trillion public tokens—four times more code data than its predecessor—and post-trained via supervised fine-tuning, rejection sampling, PPO, and DPO over more than ten million human-annotated examples. The 8B instruct model achieves 68.4 on MMLU, 62.2 on HumanEval, and 79.6 on GSM8K, establishing it as the strongest reasoning baseline among the three models evaluated here. Its primary limitation for the present task is that none of these benchmarks directly measures long-form narrative compression.

Qwen2-1.5B-Instruct (Hui et al., 2024) occupies a third point in the design space: a compact 1.54-billion-parameter model with SwiGLU activations, attention QKV bias, GQA, and a multilingual tokenizer covering English, Chinese, and 27 additional languages, post-trained with supervised fine-tuning and DPO. Despite its reduced capacity, the model shows substantial benchmark gains over its predecessor—MMLU from 43.7 to 52.4, GSM8K from 35.3 to 61.6—demonstrating that careful multilingual data curation and alignment can meaningfully compensate for parameter count. Its inclusion in this study tests whether a compact instruction-tuned model can produce competitive summaries when the task requires natural language generation rather than reasoning.

A critical observation shared across all three LLMs is that their primary evaluation suites emphasize reasoning, mathematics, coding, and dialogue—not long-form literary summarization. This shared gap in official benchmarking is precisely the regime that BookSum occupies, meaning the transfer performance of these models to chapter-length narrative compression is an open empirical question that the present study addresses directly. Furthermore, all three models are evaluated here as off-the-shelf instruction-tuned systems without fine-tuning on BookSum, which isolates the effect of general-purpose instruction following on this specialized task.

2.4 Research Gap and Contribution of the Present Study

The foregoing review identifies three specific gaps that the present study addresses. First, no prior work has directly compared Mistral-7B-Instruct-v0.1, Meta-Llama-3-8B-Instruct, and Qwen2-1.5B-Instruct against a common extractive baseline on the chapter-level split of BookSum. Second, the structural limitations of embedding-based extractive methods—inability to paraphrase, truncation-induced coverage loss, and insensitivity to narrative structure—have not been quantitatively demonstrated against these specific LLMs on this dataset. Third, the source-level variance in BookSum reference lengths has not previously been foregrounded as a confound in cross-model ROUGE comparisons. The present study contributes empirical results on all three fronts, using a standardized preprocessing pipeline across all systems, with BERT-compatible chunking used for the extractive baseline.

3 Method

3.1 Dataset

The dataset used in this study is BookSum (Krzyściński et al., 2021), a large-scale benchmark specifically designed for long-document summarization of literary texts. BookSum was introduced by Kryściński et al. (2022) and is publicly available on the Hugging Face Hub (Lhoest et al., 2021) under the identifier kmfoda/booksum with a BSD-3-Clause license.

Data source and collection. The dataset aggregates human-written summaries from eight online literary analysis platforms: CliffNotes, GradeSaver, Shmoop, SparkNotes, NovelGuide, PinkMonkey, TheBestNotes, and BookWolf (Krzyściński et al., 2021). These platforms provide professionally written chapter-level summaries for a wide range of canonical literary works, making them a reliable proxy for high-quality reference summaries. The underlying book chapters were sourced from Project Gutenberg (Project Gutenberg, 1971), which provides public-domain literary texts. As described by Kryściński et al. (Krzyściński et al., 2021), the collection process involved crawling these literary analysis platforms and aligning each chapter-level summary with its corresponding chapter text, followed by deduplication and quality filtering. Each entry in the dataset pairs a raw book chapter with one or more corresponding summaries drawn

from these analysis sources. The dataset was collected and released in 2021 (Krzyściński et al., 2021).

Dataset splits and statistics. BookSum is divided into three standard splits. After applying our preprocessing pipeline, the usable splits consist of 7,990 training samples, 1,326 validation samples, and 1,174 test samples. All experiments in this paper are conducted on the 1,174 test samples.

Data characteristics. Each sample in the preprocessed dataset contains the following fields: `chapter_clean` (the preprocessed chapter text used as model input), `summary_clean` (the preprocessed reference summary used for evaluation), `chapter_words_clean` and `summary_words_clean` (word counts for the chapter and summary respectively), `compression_ratio` (the ratio of summary length to chapter length), `chunks_bert` (the chapter text segmented into BERT-compatible chunks of 350 words with a 50-word overlap (Devlin et al., 2019)), and `num_chunks_bert` (the number of chunks per chapter).

3.2 Method Description

This study evaluates two summarization paradigms: an extractive baseline and abstractive instruction-tuned large language models. The overall pipeline consists of preprocessing the BookSum chapters, segmenting long documents into manageable chunks, generating summaries under controlled prompt conditions, and evaluating the outputs using automatic metrics and statistical tests.

3.2.1 Preprocessing and Chunking

Before summarization, all BookSum splits were processed using the same cleaning pipeline. First, aggregate rows were removed because they combine multiple chapters and do not match the chapter-level focus of this study. Rows with missing chapter or summary text were then checked, followed by the removal of empty strings after whitespace stripping. The remaining chapter and summary texts were normalized by applying Unicode normalization, fixing repeated whitespace, and removing control characters. We then computed cleaned word counts and removed samples with chapters shorter than 100 words or summaries shorter than 20 words. After preprocessing, the final usable splits contained 7,990 training samples, 1,326 validation samples, and 1,174 test samples.

Since most BookSum chapters exceed the

input limits of encoder-based models, we applied sentence-aware hierarchical chunking to the cleaned chapters across all three splits. Each chunk contains up to 350 words, with a 50-word overlap between consecutive chunks to preserve local context across boundaries. The chunking function respects sentence boundaries to avoid splitting sentences in the middle. In the training split, each chapter produced 10.1 chunks on average, with a minimum of 1 chunk and a maximum of 100 chunks. The resulting chunks were stored in the `chunks_bert` field and used in later summarization experiments.

3.2.2 Extractive Baseline

For the extractive baseline, we implemented a DistilBERT-based sentence selection approach using `distilbert-base-uncased`. This model does not generate new text; instead, it selects the most representative sentences from the source. During evaluation, the first preprocessed BERT-compatible chunk of each chapter was used as input. The chunk was split into individual sentences, and each sentence was encoded using DistilBERT. A mean embedding was then computed to represent the overall semantic content of the chunk. Sentences were ranked according to cosine similarity with this mean embedding, and the top three sentences were selected and reordered according to their original position in the chapter. This produced an extractive summary composed entirely of sentences copied from the source text.

This baseline provides a comparison point for assessing the extent to which abstractive LLMs improve over direct sentence extraction.

3.2.3 Abstractive LLM Summarization

For abstractive summarization, we evaluated three instruction-tuned large language models: `NousResearch/Meta-Llama-3-8B-Instruct`, `mistralai/Mistral-7B-Instruct-v0.1`, and `Qwen/Qwen2-1.5B-Instruct`. These models were used in a zero-shot setting without fine-tuning on BookSum. Each model was prompted to generate a summary of approximately 200 words for each chapter. Input chapters were truncated to the first 3,000 characters prior to prompt construction.

All LLMs were evaluated on the full BookSum test set of 1,174 samples. The generation process used deterministic decoding with `do_sample=False` and `max_new_tokens=200`.

For chat-based models, prompts were formatted using each model’s chat template. This ensured that the models were evaluated under their intended instruction-following format while keeping the task and prompt structure consistent across models.

3.2.4 Bias Prompt Conditions

To examine whether demographic framing affects summarization behavior, we designed seven controlled prompt conditions. The baseline condition contained no demographic cue and asked the model to summarize the chapter directly. The six bias conditions added a short framing sentence before the summarization instruction, describing the author using one demographic attribute related to age, gender, or literacy level. The seven conditions were:

- `baseline`: no demographic cue.
- `ageism_young`: the chapter is framed as written by a young author.
- `ageism_old`: the chapter is framed as written by a senior author.
- `gender_male`: the chapter is framed as written by a male author.
- `gender_female`: the chapter is framed as written by a female author.
- `literacy_high`: the chapter is framed as written by an author with a high education level.
- `literacy_low`: the chapter is framed as written by an author with a basic education level.

This setup allows a direct comparison between the neutral baseline prompt and each demographic condition while keeping the chapter content and summarization instruction constant.

3.2.5 Evaluation Metrics

We evaluated the generated summaries using ROUGE-1, ROUGE-2, ROUGE-L, BERTScore F1, and extractive ratio. ROUGE-1 and ROUGE-2 measure unigram and bigram overlap with the reference summary, while ROUGE-L captures the longest common subsequence between the generated and reference summaries. BERTScore F1 measures semantic similarity using contextual embeddings, making it useful for evaluating abstractive outputs that may paraphrase the reference rather than copy its wording.

In addition, we report an extractive ratio based on sentence-level overlap. For the DistilBERT extractive baseline, this ratio reflects direct source copying because the generated summary is constructed from sentences selected from the input text. For the LLM outputs, the implemented ratio measures sentence-level overlap with the reference summaries. Therefore, near-zero values for the LLMs should be interpreted cautiously as a proxy for abstractive behavior.

3.2.6 Statistical Analysis

For the bias evaluation, each demographic condition was compared against the baseline condition using paired statistical analysis. We used the Wilcoxon signed-rank test to determine whether the score differences between each bias condition and the baseline were statistically significant. Since statistical significance can be inflated by large sample sizes, we also computed Cohen’s d effect size to assess practical significance. Following common interpretation thresholds, values of $|d| < 0.2$ were treated as negligible. Finally, 95% confidence intervals were examined to determine whether the observed differences were stable across conditions.

4 Results

This section presents the experimental results from two perspectives. First, we compare the extractive baseline with the three abstractive LLMs under the neutral baseline prompt. Second, we examine whether demographic prompt conditions lead to measurable changes in summarization quality.

4.1 Overall Summarization Performance

Table 1 reports the average ROUGE-1, ROUGE-2, ROUGE-L, BERTScore F1, and extractive ratio for the DistilBERT-based extractive baseline and the three abstractive LLMs. All models were evaluated on the full BookSum test set of 1,174 samples, with the LLMs evaluated under the neutral baseline prompt condition.

Figure 1 visually confirms that the abstractive LLMs outperform the DistilBERT extractive baseline across the main evaluation metrics.

Across all evaluation metrics, the abstractive LLMs outperform the extractive baseline by a clear margin. LLaMA 3-8B-Instruct achieves the highest performance, with ROUGE-1 = 0.3376, ROUGE-2 = 0.0620, ROUGE-L = 0.1683, and BERTScore F1 = 0.7763. Mistral-7B-Instruct ranks second across

all metrics, followed by Qwen2-1.5B-Instruct. Although Qwen2 is substantially smaller than the other two LLMs, it still achieves a much higher ROUGE-1 score than the extractive baseline.

The gap between the extractive and abstractive systems is especially visible in ROUGE-1 and BERTScore F1. The DistilBERT extractive baseline obtains ROUGE-1 = 0.1904 and BERTScore F1 = 0.7163, while the weakest abstractive model, Qwen2, achieves ROUGE-1 = 0.2854 and BERTScore F1 = 0.7630. This indicates that the LLM-generated summaries are closer to the human-written references both lexically and semantically.

The extractive ratio further confirms the difference between the two summarization paradigms. The DistilBERT baseline has an extractive ratio of 1.0000, meaning that its summaries consist entirely of sentences copied from the source text. In contrast, the LLMs show near-zero sentence-level overlap with the reference summaries. Since this overlap was computed against the references, it should be interpreted cautiously as a proxy for abstractive behavior.

4.2 Bias Evaluation Across Prompt Conditions

Table 2 reports the average ROUGE-1 and BERTScore F1 for each demographic prompt condition across the three LLMs. The conditions include the neutral baseline prompt and six demographic framing prompts related to age, gender, and literacy level.

Figure 2 shows that ROUGE-1 scores remain stable across demographic prompt conditions.

Overall, the scores remain highly stable across all demographic prompt conditions. For LLaMA 3, ROUGE-1 ranges from 0.3359 under the `literacy_high` condition to 0.3465 under the `literacy_low` condition, producing a maximum spread of 0.0106. For Mistral, ROUGE-1 ranges from 0.3132 to 0.3192, with a maximum spread of 0.0060. Qwen2 shows the smallest variation, with ROUGE-1 ranging from 0.2815 to 0.2861, a spread of only 0.0046.

BERTScore F1 follows the same pattern. The differences across conditions are small for all three models, suggesting that demographic framing does not substantially change the semantic similarity between generated summaries and reference summaries. Importantly, the baseline condition remains among the strongest conditions for Mistral and

Table 1: Overall summarization performance. All models (DistilBERT extractive baseline and LLMs) are evaluated on the same 1,174 test samples. All BERTScore values use distilbert-base-uncased. For the DistilBERT baseline, an extractive ratio of 1.0 indicates that all output sentences are copied directly from the source. For the LLMs, the reported ratio reflects sentence-level overlap with the reference summaries.

Model	R-1	R-2	R-L	BERTScore F1	Extr. Ratio
DistilBERT Extractive	0.1904	0.0205	0.1055	0.7163	1.0000
Qwen2-1.5B-Instruct	0.2854	0.0441	0.1503	0.7630	0.0000
Mistral-7B-Instruct	0.3192	0.0577	0.1645	0.7747	0.0001
LLaMA 3-8B-Instruct	0.3376	0.0620	0.1683	0.7763	0.0000

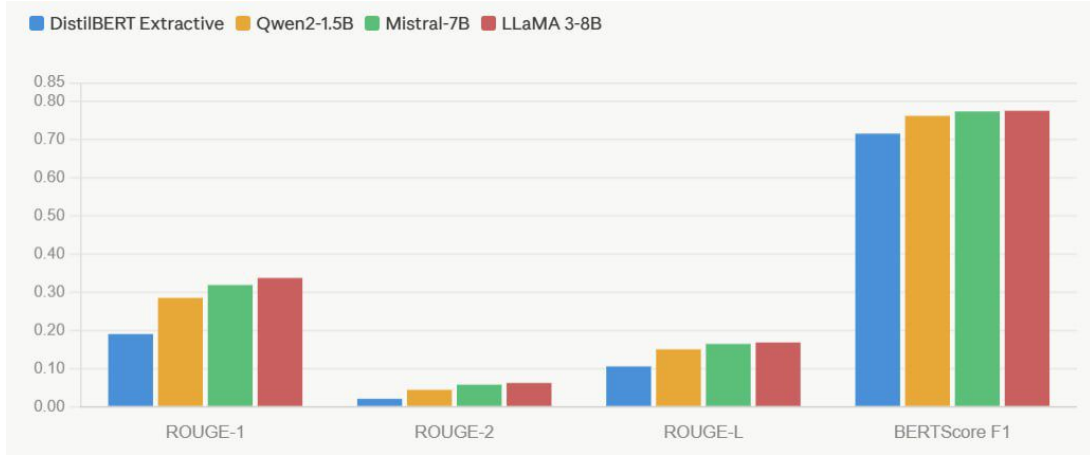


Figure 1: Comparison of ROUGE and BERTScore results across the extractive baseline and the three abstractive LLMs.

Qwen2, while LLaMA 3’s highest ROUGE-1 under `literacy_low` is not accompanied by a corresponding improvement in BERTScore F1. This suggests that the small metric fluctuations do not reflect a consistent demographic effect.

4.3 Statistical Significance and Effect Size

To determine whether the observed differences between demographic conditions and the baseline are meaningful, we conducted paired statistical testing using the Wilcoxon signed-rank test. Table 3 reports the Wilcoxon p -values and Cohen’s d effect sizes for ROUGE-1 comparisons between each demographic condition and the baseline.

The Wilcoxon test identifies some statistically significant differences. For LLaMA 3, only the `literacy_low` condition is statistically significant. For Mistral, five conditions are significant: `ageism_young`, `ageism_old`, `gender_female`, `literacy_high`, and `literacy_low`. For Qwen2, two conditions are significant: `ageism_old` and `literacy_high`.

However, statistical significance alone is not sufficient to indicate bias. Since the LLM experiments use 1,174 paired samples, even very small differences can produce significant p -values. Cohen’s d

shows that all observed differences are negligible in practical terms. The largest effect size is $d = -0.129$ for LLaMA 3 under the `literacy_low` condition, which remains below the conventional threshold of $|d| < 0.2$ for a small effect. Mistral’s largest effect size is $d = 0.083$, and Qwen2’s largest effect size is $d = 0.055$. These values indicate that the demographic prompt conditions do not produce practically meaningful changes in summarization performance.

4.4 Confidence Interval Analysis

In addition to statistical testing and effect-size analysis, we examined 95% confidence intervals for the mean ROUGE-1 scores of each condition. Most demographic conditions showed substantial overlap with the corresponding baseline intervals. The main exception was LLaMA 3 under the `literacy_low` condition, where ROUGE-1 showed a statistically detectable difference. However, the associated Cohen’s d remained negligible, and BERTScore did not show a corresponding improvement. Therefore, the confidence interval analysis supports the conclusion that the observed differences are not practically meaningful.

Table 2: Average ROUGE-1 and BERTScore F1 per bias condition for each LLM. Bold indicates the highest ROUGE-1 score for each model.

Condition	LLaMA 3		Mistral		Qwen2	
	R-1	BSc	R-1	BSc	R-1	BSc
baseline	0.3376	0.7763	0.3192	0.7747	0.2854	0.7630
ageism_young	0.3379	0.7750	0.3154	0.7720	0.2861	0.7617
ageism_old	0.3377	0.7750	0.3132	0.7715	0.2823	0.7619
gender_male	0.3384	0.7755	0.3176	0.7726	0.2843	0.7613
gender_female	0.3366	0.7754	0.3163	0.7719	0.2850	0.7613
literacy_high	0.3359	0.7686	0.3161	0.7724	0.2815	0.7615
literacy_low	0.3465	0.7703	0.3159	0.7725	0.2828	0.7613

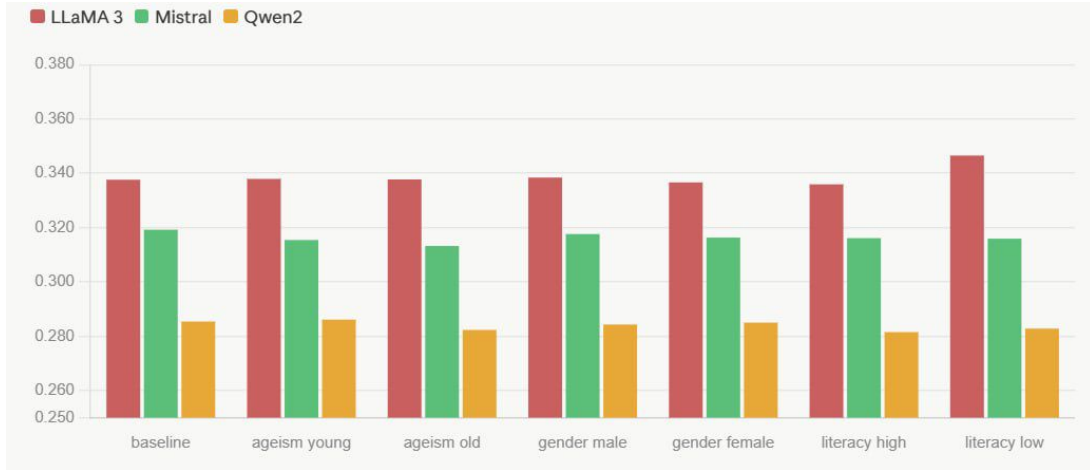


Figure 2: ROUGE-1 scores across demographic prompt conditions for the three LLMs.

4.5 Qualitative Output Comparison

A qualitative inspection of generated summaries further supports the quantitative findings. Across the seven prompt conditions, the summaries generated by each model generally preserve the same core chapter events, characters, and narrative focus. Differences between conditions are mostly limited to wording, sentence structure, or minor changes in emphasis rather than systematic changes in content quality.

The LLM outputs appear qualitatively more abstractive than the extractive baseline, as they generally use compressed and reformulated wording rather than copying selected source sentences. This observation is consistent with their low sentence-level overlap with the reference summaries, although the overlap metric should be interpreted cautiously because it is computed against the references. Overall, the qualitative comparison confirms that the demographic framing prompts do not lead to consistent changes in the generated summaries, and that the main distinction in output behavior is between extractive copying and abstractive paraphrasing.

5 Discussion

5.1 Extractive vs. Abstractive Performance

The quantitative results confirm the hypothesis stated in Section 2: embedding-based extractive summarization is structurally ill-suited to long-form literary chapters of the kind found in BookSum. The DistilBERT extractive baseline achieved ROUGE-1 = 0.1904 and BERTScore F1 = 0.7163 on the test set (1,174 samples), while all three abstractive LLMs substantially outperformed these scores on the full 1,174-sample test set. Under the neutral baseline prompt, LLaMA 3-8B-Instruct reached ROUGE-1 = 0.3376, Mistral-7B-Instruct reached ROUGE-1 = 0.3192, and Qwen2-1.5B-Instruct reached ROUGE-1 = 0.2854. The gap between the extractive baseline and the best abstractive model on ROUGE-1 is approximately 0.147 points, representing a substantial relative improvement.

This result is attributable to the structural limitations of the extractive approach. Because the baseline selects only three sentences from the first BERT-compatible chunk of the chapter, the majority of each chapter’s content is unrepresented. Literary chapters in BookSum have a median length

Table 3: Wilcoxon signed-rank test p -values and Cohen’s d effect sizes for ROUGE-1 comparisons between each demographic condition and the baseline. * $p < .05$. All Cohen’s d values are negligible ($|d| < 0.2$).

Condition	LLaMA 3		Mistral		Qwen2	
	p	d	p	d	p	d
ageism_young	0.5104	−0.002	0.0071*	0.052	0.4536	−0.009
ageism_old	0.7119	−0.002	<0.0001*	0.083	0.0129*	0.044
gender_male	0.2771	−0.012	0.3628	0.022	0.8391	0.016
gender_female	0.2163	0.015	0.0261*	0.040	0.8478	0.006
literacy_high	0.3318	0.024	0.0018*	0.043	0.0147*	0.055
literacy_low	<0.001*	−0.129	0.0044*	0.046	0.1258	0.036

of 2,934 words, meaning the baseline examines at most a small fraction of the source. Even within that fragment, cosine similarity to a mean document embedding favours lexically central sentences over narratively important ones—a known weakness of centrality-based extraction for fiction and literary prose (Reimers and Gurevych, 2019). The abstractive LLMs, by contrast, generate novel text from the truncated chapter prefix used in the prompt. Although this input does not cover the full chapter, the models still produce summaries that align more closely with the human-written references than the extractive baseline under the evaluated setup. The extractive ratio of 1.0000 for the DistilBERT baseline further illustrates its fully extractive behavior. The near-zero overlap values for the LLMs suggest that their outputs rarely reproduce complete reference-summary sentences; however, this metric does not directly measure copying from the source chapter.

Among the three LLMs, LLaMA 3 achieves the highest scores, followed by Mistral and then Qwen2. This ordering is broadly consistent with parameter count: LLaMA 3-8B and Mistral-7B are comparable in size, while Qwen2-1.5B is substantially smaller. Notably, Qwen2 still outperforms the extractive baseline by a large margin despite having fewer than one-fifth of the parameters of the other two models, suggesting that general-purpose instruction tuning provides strong zero-shot summarization capability even at compact scale.

5.2 Demographic Bias Analysis

The bias evaluation yields a clear and consistent finding across all three LLMs: demographic prompt cues have no practically meaningful effect on summarization quality. This conclusion is supported by three complementary lines of evidence.

Score magnitudes. Across all seven conditions and all three models, the range of ROUGE-1 scores is extremely narrow. For LLaMA 3, the maximum

spread is 0.0106 points; for Mistral, 0.0060; and for Qwen2, 0.0046. BERTScore F1 shows similarly narrow variation, with the largest within-model spread being 0.0077 for LLaMA 3. The sentence-level overlap values are effectively zero for all conditions across all models, suggesting that demographic cues do not substantially affect verbatim overlap with the reference summaries.

Statistical vs. practical significance. The Wilcoxon signed-rank test, applied with $n = 1,174$ paired samples, is sensitive enough to detect differences of a few thousandths of a ROUGE point. Accordingly, some conditions reach statistical significance. However, Cohen’s d for every condition across every model falls well below the negligibility threshold of 0.2. The largest observed effect is $d = -0.129$ for LLaMA 3 (literacy_low), $d = 0.083$ for Mistral (ageism_old), and $d = 0.055$ for Qwen2 (literacy_high). These values indicate that the statistically significant results are artefacts of statistical power at large n , not evidence of practically meaningful differences.

Confidence intervals. Most 95% confidence intervals for mean ROUGE-1 overlapped substantially with the corresponding baseline intervals. The main exception was LLaMA 3 under the literacy_low condition; however, its effect size remained negligible, so the difference should not be interpreted as practically meaningful demographic sensitivity.

5.3 Interpretation and Implications

The absence of meaningful demographic bias across all three models is noteworthy given the diversity of their architectures and training regimes. A plausible explanation is that the demographic cue—a single prepended sentence—is overwhelmed by the much larger chapter text that immediately follows it. With chapters averaging over 4,000 words, the model’s attention is dominated by the source content, and the one-sentence

framing has negligible influence on the final output. This interpretation is also consistent with the stable ROUGE and BERTScore results across prompt conditions, which suggest that the models remain primarily driven by the chapter content rather than the demographic framing.

From a practical standpoint, these results are reassuring for the deployment of instruction-tuned LLMs in long-document summarization: the models do not produce systematically lower-quality summaries when the prompt implies that the source was authored by a person of a particular age, gender, or educational background.

6 Conclusion

This paper presented a systematic comparison of extractive and abstractive summarization methods on the BookSum long-document benchmark, evaluated on 1,174 chapter-level test samples. We implemented a DistilBERT-based centroid similarity extractive baseline alongside three state-of-the-art instruction-tuned LLMs: LLaMA 3-8B-Instruct, Mistral-7B-Instruct, and Qwen2-1.5B-Instruct. We further introduced a controlled bias evaluation framework comprising seven prompt conditions that inject demographic cues related to author age, gender, and literacy level, and assessed any resulting performance differences using ROUGE, BERTScore, the Wilcoxon signed-rank test, Cohen’s d effect size, and 95% confidence intervals.

Our experiments yield two principal findings. First, abstractive LLMs substantially outperform the extractive baseline across all evaluation metrics. This gap is attributable to the structural limitations of centrality-based extraction on long literary texts, where narrative coherence, long-range coreference, and content distribution across thousands of words cannot be addressed by selecting a handful of sentences from a single document chunk.

Second, none of the three LLMs exhibit meaningful demographic bias in response to author-framing cues. While the Wilcoxon test flagged some conditions as statistically significant, Cohen’s d effect sizes were negligible for all conditions across all models, with the largest observed value being $d = -0.129$ (LLaMA 3, literacy_low). Most confidence intervals overlapped substantially with the baseline, with the exception of LLaMA 3 under the literacy_low condition. Sentence-level overlap with the reference summaries also remained near zero. These results suggest that, under the

tested prompt-framing conditions, the models did not show practically meaningful demographic sensitivity in long-document summarization.

Limitations

This study has several limitations that should be considered when interpreting its findings.

Reference summary variance. BookSum reference summaries are drawn from multiple literary analysis platforms whose average summary lengths differ substantially—GradeSaver averages 502 words versus BookWolf’s 194 words. Because ROUGE is sensitive to reference length, scores computed on BookSum are partly a function of which source provided the reference, introducing a source-level confound that affects all models.

No fine-tuning. All three LLMs were evaluated as off-the-shelf instruction-tuned systems without any fine-tuning on BookSum. While this isolates the zero-shot capability of each model, it means the results do not reflect the full potential of these architectures on this task.

Prompt-level bias evaluation only. The bias framework manipulates only the framing sentence prepended to the chapter text. It does not test bias arising from the content of the chapters themselves or from more subtle prompt constructions. Future work could examine whether bias emerges under adversarially constructed prompts or with demographic cues embedded in longer system prompts.

Metric coverage. ROUGE and BERTScore measure lexical and semantic overlap with a reference summary, but do not assess factual consistency, narrative fidelity, or the specific framing choices made by the model. A model could produce summaries that subtly differ in emphasis or perspective across demographic conditions without those differences being captured by the metrics used here.

Single-language evaluation. All experiments were conducted on English-language literary texts. Given Qwen2’s multilingual design, evaluating bias and summarization quality across other languages remains an open direction for future work.

Acknowledgments

This work was conducted as part of the IT469 Human Language Technologies course at King Saud University.

References

- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, pages 4171–4186.
- Abhimanyu Dubey and 1 others. 2024. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.
- Isabel O. Gallegos and 1 others. 2024. Bias and fairness in large language models: A survey. *arXiv preprint arXiv:2309.00770*.
- Binyuan Hui, Jian Yang, Zeyu Cui, Jiaxi Yang, Dayiheng Liu, Lei Zhang, and 1 others. 2024. Qwen2 technical report. *arXiv preprint arXiv:2407.10671*.
- Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, L  lio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timoth  e Lacroix, and William El Sayed. 2023. Mistral 7B. *arXiv preprint arXiv:2310.06825*.
- Wojciech Kryscinski, Nazneen Rajani, Divyansh Agarwal, Caiming Xiong, and Dragomir Radev. 2022. Booksum: A collection of datasets for long-form narrative summarization. *arXiv preprint arXiv:2105.08209*.
- Wojciech Krzy  ci  ski, Nazneen Rajani, Divyansh Agarwal, Caiming Xiong, and Dragomir Radev. 2021. Booksum: A collection of datasets for long-form narrative summarization. *arXiv preprint arXiv:2105.08209*.
- Quentin Lhoest and 1 others. 2021. Datasets: A community library for natural language processing. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 175–184. Association for Computational Linguistics.
- Project Gutenberg. 1971. [Project gutenberg](#).
- Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of EMNLP-IJCNLP*, pages 3982–3992.
- Wenhui Wang, Furu Wei, Li Dong, Hangbo Bao, Nan Yang, and Ming Zhou. 2020. MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers. *Advances in Neural Information Processing Systems*.